DiffPR-Net: Few-Shot Remote Sensing Scene Classification Based on Generative Diffusion and Prototype Rectified Model

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***Abstract*— Few-shot remote sensing scene classification (FSRSSC) aims to identify unseen scene classes from limited labeled samples, facing the challenge of accurately modeling data distribution and preserving image details in complex backgrounds with high intraclass variance and interclass similarity. To address this challenge, we propose a novel diffusion prototype rectified network (DiffPR-Net), which is comprised of three core modules: diffusion augmentation (DA), dual attention fusion module (DAFM), and prototype rectified module (PRM). The DA is constructed to generate high-quality remote sensing images with the objective of augmenting the training dataset. Besides, the DAFM facilitates the model to focus discriminative regions by transmitting highly fused image detail features from higher to lower layers. What is more, the PRM addresses prototype deviation by adaptively assigning temporary labels to unlabeled data based on prediction confidence, thereby correcting the initial prototypes. Experiments indicate that our proposed method is highly promising, achieving competitive or state-of-the-art (SOTA) classification performance while addressing the scarcity of remotely sensed data and enhancing focus on discriminative regions.**

***Index Terms*— Data augmentation, diffusion model, few-shot remote sensing scene classification (FSRSSC), prototype network.**

I. Introduction

FEW-SHOT remote sensing scene classification (FSRSSC) is the process of assigning a specific semantic label to each unlabeled remote sensing image based on its semantic content, using multispectral or hyperspectral image data [1]. It has practical applications in urban planning, environmental monitoring [2], agricultural development [3], disaster monitoring [4], and land resource management [5].

Deep learning methods have become popular in classification task due to their excellent feature extraction capabilities [6]. However, these methods often require a large number of labeled samples for training. If there are insufficient labeled samples available, the deep learning model may overfit, resulting in a decrease in performance [7]. Labeling remote sensing images is a time-consuming and laborious task that requires experts with extensive knowledge of the remote sensing domain. Few-shot learning (FSL) was proposed by Fei-Fei et al. [8] to mimic human reasoning and learning from limited data, inspired by the ability of human children to learn new concepts quickly from a very small number of examples. In recent years, researchers introduce FSL into remote sensing scene classification task to reduce the burden of labeled data, which leads to the task of FSRSSC gain widespread attention.

To address the limited nature of the data, data augmentation is applied with the generation of new samples to increase the amount and variety of data available. Xu et al. [9] developed a generative adversarial network (GAN)-based method using the scaled exponential linear units (SELUs) activation to generate remote sensing images with a significant reduction in computation time. Guo et al. [10] further explored a semi-supervised GAN-based approach where labeled and unlabeled real samples as well as GAN-based generated samples are fed into the network for classification. However, the gradient may disappear or explode if not designed properly, resulting in poor quality of the generated remote sensing images. Besides, the small interclass and large intraclass variance is another challenge in FSRSSC task, where small interclass is visually indistinguishable despite being distinct classes and large intraclass variance is that samples of the same semantic category frequently display notable visual differences, as is shown in Fig. 1.

In this article, we propose a diffusion-based generative model with the application of meta-learning, named diffusion prototype rectified network (DiffPR-Net), which can generate diverse support set samples and enhance feature representation. Diffusion augmentation (DA) employs forward diffusion and backward denoising processes on the support set, which can learn the latent distribution of input data and generate new samples similar to the training data to solve the scarcity of labeled samples in remote sensing. Besides, the dual attention

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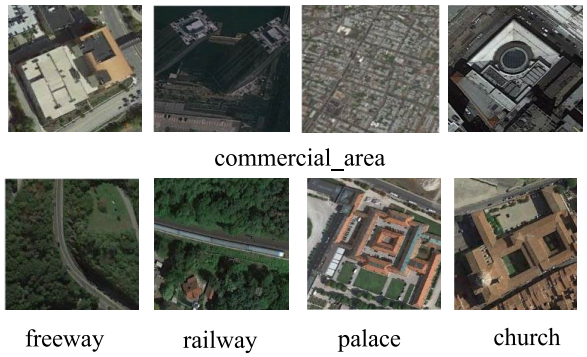


Fig. 1. Illustrations of large intraclass and small interclass variances, where the top row is large intraclass variance in the category of commercial_area, and bottom row is small interclass of among freeway\railway and palace\church.

fusion module (DAFM) integrates global feature attention and spatial attention on local features, thereby fusing representative and discriminative key features. By embedding the DAFM within each network layer of ResNet-12 and iteratively fusing features across network layers, high-quality image details fused at higher layers are propagated to lower layers for enhanced feature extraction. The prototype rectified module (PRM) enhances the support set through the use of a pseudo-labeling strategy, which assigns temporary labels to unlabeled data based on adaptive prediction confidence. Additionally, it dynamically allocates relationship weights to refine the initial prototypes, thereby narrowing intraclass distances and enlarging interclass distances. The contributions of this work can be summarized as follows.

1) We propose a novel methodology employing DA to address the challenge of data scarcity in remote sensing scene classification. This research represents a pioneering exploration of diffusion models within the domain of FSRSSC.
2) A DAFM was constructed to adapt to the high intraclass variance and low interclass variance in remote sensing images. The module obtains high-quality image features by fusing and transferring global and local features of intra-layer channels.
3) The PRM is proposed to address the issue of prototype bias in traditional prototype, which is accomplished by using a pseudo-labeling strategy to generate more representative and discriminative class-level prototypes for rectifying the prototype.

The article is structured as follows: Section II outlines the work related to FSRSSC. Section III provides a detailed description of the proposed DiffPR-Net. Section IV presents the extensive experimental results and analyses on three public remote sensing datasets. Finally, Section V summarizes the conclusions.

II. Related Work

A. Few-Shot Remote Sensing Scene Classification

FSRSSC is a crucial research area in remote sensing image processing. Its objective is to identify unlabeled samples from new categories using a priori knowledge obtained from few shot labeled samples. Existing FSRSSC methods are typically classified into two categories: transfer learning-based methods [11], [12], [13] and meta-learning-based methods as in [14], [15], [16], [17], [18], [19], [20], and [21].

Xing et al. [11] proposed the decision fusion method (DFM) to address the negative migration problem between pre-trained and remotely sensed data caused by data discrepancy. The DFM suggests using two pre-trained models to perform remote sensing classification separately and then fusing the decisions to obtain the final classification results. Gong et al. [12] proposed the data architecture network (DANet) to address the FSRSSC problem from both data and architecture perspectives. Specifically, the authors designed a quad-patch data augmentation strategy and a two-path aggregated attention module to focus on the key features of remote sensing images. In their work, Ji et al. [13] utilized multiple auxiliary tasks to assist in classification for the FSRSSC task and introduced regularization techniques to further improve the generalization of the model. Although transfer learning has shown promising results, it may require additional training time and data to adapt to the target domain's features, which can limit its performance in few shot conditions. This article addresses the issue of large differences between the source and target domains in transfer learning by using a meta-learning-based approach, which can lead to performance degradation.

Li et al. [14] proposed a discriminative learning method called adaptive match network (DLA-MatchNet) within the meta-learning framework, which uses an attention mechanism and an adaptive distance matcher to measure the similarity between support and query samples. Cheng et al. [15] proposed a Siamese-prototype network (SPNet) with prototype self-calibration and mutual calibration to obtain more accurate prototypes through the interaction of information between the support samples and the query samples. Ma et al. [16] put forth a novel dictionary learning (DL) algorithm that enhances feature discriminability through the fine-tuning of pre-trained natural image feature extractors and the design of a kernel space classifier. Jia et al. [17] introduced the hard sample learning (HSL) and multiview integration (MI) network (HSL-MINet), which was designed to enhance classification performance by addressing challenging samples through spatial rotation insensitivity and high distribution uncertainty. Tian et al. [18] proposed hierarchical-relation network (HiReNet), an innovative approach that integrates a hierarchical relationship learning structure and a feature aggregation module. This method enhances metric representation learning by leveraging the hierarchical features of both query and support samples. Wang et al. [19] proposed a multiscale feature matching network (MSFMNet) for the purpose of constructing intrinsic correlations in remote sensing images and enhancing the discriminability of few-shot samples. Xu et al. [20] employed an attention-based feature optimization (ABFO) module to align and enhance target image features, thereby highlighting the target regions. In the field of meta-learning, conventional prototypical networks [21] predominantly utilize a simple averaging technique to generate prototypes. However, this approach may lead to prototype bias. To tackle this issue, this article proposes the PRM, which dynamically allocates

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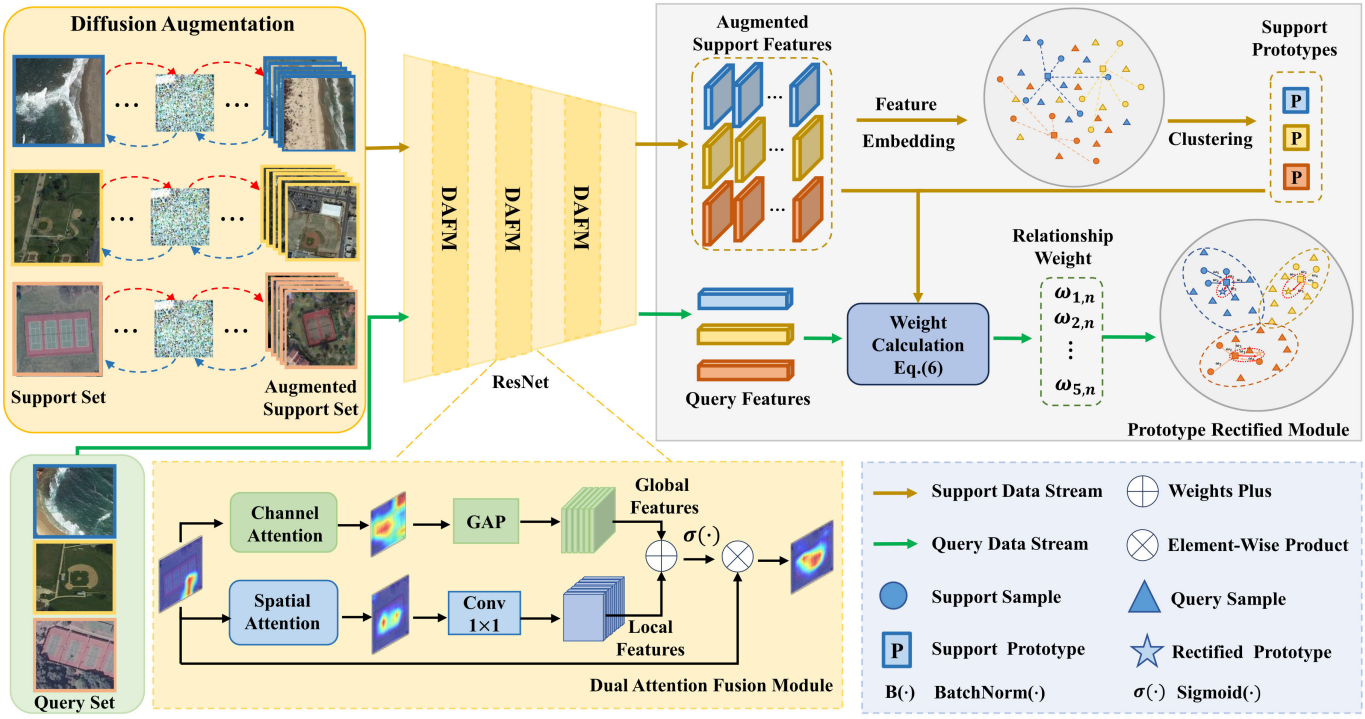


Fig. 2. Framework of our proposed DiffPR-Net, where DA is employed to simulate latent variations present in remote sensing scenes, thereby generating numerous synthetic remote sensing images corresponding to the input support set. Within each layer of ResNet-12, DAFM facilitates the transmission of fused image details from higher layers to lower layers, thereby enabling the extraction of pivotal features. PRM addresses the issue of prototype bias by employing pseudo-labeling strategies, whereby the weighted sum of original support samples and query samples is calculated to refine feature representations.

relational weights for prototype calibration, resulting in more accurate class prototypes.

B. Applications of Diffusion Models in Remote Sensing

In the field of remote sensing image processing, diffusion models, which are mathematical frameworks based on the propagation and evolution of image information, offer a number of unique advantages and significant potential for a variety of applications. In tasks such as super-resolution, semantic segmentation, and data augmentation, diffusion models can significantly enhance image quality and accuracy. Furthermore, they provide a richer and more reliable foundation for subsequent image analysis and applications. Xiao et al. [22] introduced an efficient diffusion model, EDiffSR, for remote sensing image super-resolution (RSI-SR). This model substantially reduces the computational budget while improving the detail and clarity of remote sensing images. An et al. [23] introduced a lightweight diffusion model (LWTDM) for RSI-SR. This model simplifies the denoising network and enhances model efficiency. In a study published by Ayala et al. [24], the potential of diffusion models for semantic segmentation tasks in remote sensing was investigated, offering valuable insights. Diffusion models have emerged as a significant analytical tool in recent years, with applications in data augmentation [25] garnering widespread attention. Rombach et al. [26] proposed the latent diffusion model (LDM), which encodes image features into latent data and conducts forward and reverse diffusion processes in the latent space. Diffusion models have made significant progress in image generation, outperforming GANs in terms of performance. In this study, we introduce DiffPR-Net, a network based on diffusion models, for generating high-quality remote sensing images. This enhances the accuracy of remote sensing scene classification under few-shot conditions.

III. Methodology

In addressing the constraints of remote sensing scene classification data, coupled with the challenges posed by minimal interclass variance and substantial intraclass variance, we propose DiffPR-Net, which consists of DA, DAFM, and PRM, as is shown in Fig. 2. The DA utilizes the diffusion model to enhance the data, where adds and reduces Gaussian noise to the original support set image features to generate labeled samples for support set enhancement. The DAFM is employed to transfer the global and local features of the intra-layer channels fused together, and to transfer the detailed features of the higher layers to the lower layers in order to obtain high-quality remote sensing image features. The PRM can dynamically assign a weight to each query sample based on the distance of the query set sample from the initial prototype. The unlabeled query samples are assigned temporary labels based on the adaptive prediction confidence. The prototype bias of the initial prototype is solved by using the weighted sum of the original support samples and the pseudo-labeled query samples.

A. Problem Formulation

FSRSSC aims to learn from a limited number of labeled instances and then perform recognition on new, unseen

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instances, where the dataset is divided into three sets: training set D_{train}, validation set D_{val}, and test set D_{test}. To emulate novel categories that have not been previously encountered within the context of the FSL task, the categories of the training, validation, and test sets are mutually exclusive ($D_{\text{train}} \cap D_{\text{val}} \cap D_{\text{test}} = \emptyset$).

In contrast to traditional deep learning paradigms, meta-learning primarily employs meta-learning tasks instead of individual samples to train deep models, which is referred to as episodic training. The training and testing phases involve a large number of episodes, each following the classical N-way K-shot setup. In each meta-task, there are N categories, where K labeled samples from each category serve as the support set $D_s = \left\{(x_i^s, y_i^s)\right\}_{i=1}^{N\times K}$, and Q unlabeled samples serve as the query set $D_Q = \left\{(x_j^q, y_j^q)\right\}_{j=1}^{N\times Q}$. The model trains on the support set and evaluates the performance on the query set. The support set and query set have different remote sensing images ($D_s \cap D_Q = \emptyset$). Typically, K is set to 1 or 5 and N is set to 5, corresponding to the five-way one-shot and five-way five-shot settings in our article.

B. Diffusion Augmentation

In the field of FSRSSC, constructing classification models that are both robust and accurate presents a significant challenge due to the limited availability of labeled data. To address this challenge and improve the models' generalization ability, existing approaches consider various data augmentation techniques to expand the training set. Traditional data augmentation methods typically modify existing images through simple geometric transformations or noise addition to generate variants. With the development of image generation, GANs are applied for data augmentation, while the samples are always low quality and may encounter pattern collapse and distortion. To address these difficulties, DA relies on simulating a diffusion process, introducing iterative minor changes to generate new samples from initial data points. What is more, the embedded vectors are obtained for the incoming support set to reduce the performance gap between synthetic and real data, which can capture the initial data distribution and ensures data coverage.

The DA module comprises two processes: forward diffusion and reverse denoising. During the forward diffusion process, the diffusion module receives support set remote sensing images $x_0^s \sim Q(x)$ from each meta-task and gradually adds Gaussian-distributed noise to them, as illustrated in Fig. 3. Gaussian noise is iteratively added to the samples based on the time step t, transforming them from x_0^s to $x_1^s, x_2^s, \ldots, x_T^s$. The probability of remote sensing noise samples transitioning from x_{t-1}^s to x_t^s at step t is computed as the following equation:

$$Q(x_t^s|x_{t-1}^s) = N(\sqrt{1-\beta_t}x_{t-1}^s, \beta_t I) \tag{1}$$

where x_{t-1}^s and x_t^s, respectively, denote the noise remote sensing images at time steps $t-1$ and t. $N(.)$ represents the Gaussian distribution, I denotes the standard normal distribution, and $\{\beta_t\}_{t=0}^T \in (0, 1)$ is the diffusion rate at time step t,

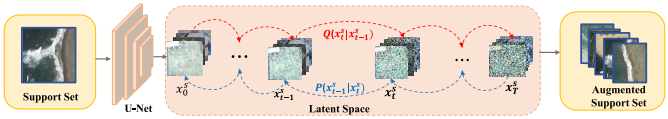


Fig. 3. Overview of DA, where the support set samples introduced in the meta-task undergo a process whereby, based on the time step t, random noise is added to the extracted original sample features. Concurrently, the input class names are subjected to text encoding. In conclusion, new labeled support set samples are synthesized through a procedure involving random denoising encoding.

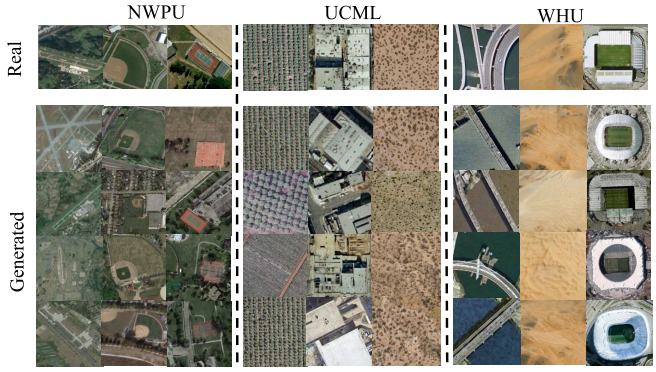


Fig. 4. Examples visualization of real images and synthesized remote sensing images by DA across three datasets.

which can be learned or kept constant. Here, $(1-\beta_t)^{1/2}x_{t-1}^s$ and $\beta_t I$ represent the mean and variance of $Q(x_t^s|x_{t-1}^s)$.

In the process of inverse remote sensing image denoising, the training diffusion module network gradually denoises the noise-added remote sensing image instances to obtain the original remote sensing image instances x_0^s. At the tth step of the inverse denoising process, the denoising process is applied to the remote sensing image instance x_t^s with Gaussian noise to obtain the remote sensing instances x_{t-1}^s from the $t-1$ step. Given x_t^s, the conditional probability of x_{t-1}^s can be derived according to the following equation:

$$P(x_{t-1}^s|x_t^s) = N(\mu_\theta(x_t^s, t), \sigma_t^2 I) \tag{2}$$

where $\mu_\theta(x_t^s, t)$ and σ_t^2 represent the mean and variance of $P(x_{t-1}^s|x_t^s)$. The augmented support set D_{aug} is formed by mixing the original support set remote sensing images x_i^s from the meta-task with the synthesized remote sensing images x_i^{syn} generated through the diffusion module, that is, $D_{\text{aug}} = (x_i^{\text{aug}}, y_i^{\text{aug}})$. The DiffPR-Net classifier is trained using the augmented support set remote sensing images, gradually optimizing the model parameters.

The visualized examples of DA on three datasets are shown as Fig. 4, which exhibits superior performance in data augmentation compared with real remote sensing images. The DiffPR-Net model in this article demonstrates strong generalization ability when dealing with complex datasets and can generate remote sensing images with rich categories and clearer features.

C. Dual Attention Fusion Module

The FSRSSC task is challenging due to the complexity and diversity of objects in remotely sensed scene images.

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Traditional classification methods often struggle to perform well when faced with a limited support set. The high intra-class variance, complexity, and diversity of features, as well as the interclass similarity of remote sensing scene images, make it difficult to define clear boundaries between classes. To tackle these challenges, we suggest using a DAFM that effectively captures and utilizes abundant feature information from remotely sensed scene images to improve the accuracy and robustness of FSRSSC.

The DAFM, illustrated in Fig. 5, utilizes both channel attention and spatial attention mechanisms to fully exploit the channel and spatial correlations in remote sensing image feature maps, where representational capacity is enhanced by the parallel arrangement of channel attention and spatial attention. The ResNet-12 backbone network has been enhanced by embedding the DAFM into each layer, which aims to introduce a detailed mechanism for global and local feature fusion. It transfers fused global and local image detail features from higher to lower layers, thereby achieving inter-layer feature fusion. For a given input x, the ResNet-12 backbone network initially conducts a series of convolutional and pooling operations to progressively extract feature maps $X_l \in \mathbb{R}^{C\times H\times W}$ at each layer l of the network. Here, C, H, and W, respectively, denote the number of channels, height, and width. Following the extraction of the feature map X_l from each layer, channel and spatial attention are applied in order to obtain more refined and global features. Channel attention is employed to integrate global semantic information through a global branch, whereas spatial attention is used to focus on local structures, object positions, and spatial details in the image through a local branch.

To minimize the number of model parameters, we set the parameter matrices for $\text{Conv}_1^{1\times1}$ and $\text{Conv}_2^{1\times1}$ as $M_1 \in \mathbb{R}^{c\times c/r}$ and $M_2 \in \mathbb{R}^{c/r\times c}$, respectively, where r is a constant. The parameter matrices for $\text{Conv}_1^{1\times1}$ and $\text{Conv}_2^{1\times1}$ are distinct. The global feature $G(X_l)$ and the local feature $L(X_l)$, respectively, concentrate on global semantics and local details, forming an attention mechanism that captures different aspects of the input remote sensing image. The global and local features are fused for each channel within the layer using the following equation:

$$X_l' = X_l \otimes \sigma(L(X_l) \oplus G(X_l)) \tag{3}$$

where $\otimes$ and $\oplus$ represent element-wise multiplication and element-wise addition. $\sigma(\cdot)$ signifies the sigmoid function. The fusion of global and local features within the layer enhances the model's ability to perceive and express information at different scales. This approach enables the network to effectively leverage valuable information from higher layers at lower levels, resulting in more detailed and richer features.

D. Prototype Rectified Module

Due to the inherent complexity of content in remote sensing scenes, relying solely on a limited support set of remote sensing images to determine prototypes is insufficient for accurately representing category distributions. In the FSRSSC scenario, each meta-task's category support set comprises only K samples. The initial prototypes calculated from these limited

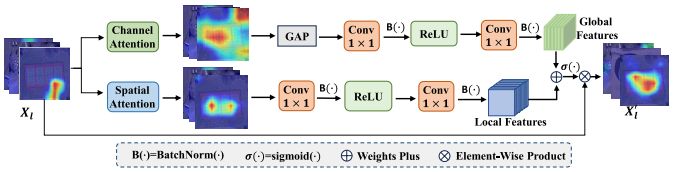


Fig. 5. Overview of DAFM, which employs parallel channels and spatial attention to extract critical information. GAP represents global average pooling and $\text{Conv}^{1\times1}$ signifies 1×1 convolutional layer. It ensures that essential features are effectively captured and integrated across different layers of the network.

support set samples exhibit deviations from the expected prototypes, thereby diminishing the accuracy of subsequent classifications. To address the issue of prototype bias, this study introduces a high-confidence query set sample to correct the initial prototypes by weighting the original support samples and query samples.

Initially, we derive the initial prototypes by straightforwardly averaging the features of the support set samples, as expressed by the following:

$$P_n^O = \frac{1}{K}\sum_{i=1}^{K} F_\theta(x_i^{\text{aug}}) \tag{4}$$

where P_n^O denotes the initial prototype for class n, x_i^{aug} represents the feature of the augmented support set sample, and $F_\theta(\cdot)$ denotes the feature extractor. We compute the Euclidean distance $D(\cdot)$ between the query set samples and the initial prototypes, and then sort them in ascending order. A smaller distance indicates higher confidence. Based on the calculated confidence, we assign temporary labels to the query set samples. We select the top Z^{top} query set samples for assigning temporary labels and add these pseudo-labeled query set samples to the augmented support set, forming an enhanced support set $D'_{\text{support}} = D_{\text{aug}} \cup D_Q^{Z^{\text{top}}}$. The weight of the confidence relationship between the enhanced support set samples and the initial prototypes is determined by the confidence level, with higher confidence samples having a larger weight proportion. This process aims to compute more accurate prototypes under a larger number of support set samples, thereby achieving the correction of the initial prototypes, which can be represented as the following equation:

$$P_n^R = \sum_{i=1}^{K+Z^{\text{top}}} \omega_{i,n} \cdot F_\theta(X'_{i,n}) \tag{5}$$

where $X'_{i,n}$ represents the feature of the enhanced support set D'_{support} sample. $\omega_{i,n}$ denotes the relationship weight between the enhanced support set sample $F_\theta(X'_{i,n})$ and the class initial prototype P_n^O, which can be obtained using the following equation:

$$\omega_{i,n} = \frac{\exp(\varepsilon \cdot \text{Cos}(F_\theta(X'_{i,n}), P_n^O))}{\sum_{j=1}^{K+Z^{\text{top}}} \exp(\varepsilon \cdot \text{Cos}(F_\theta(X'_{j,n}), P_n^O))}. \tag{6}$$

The relationship weight is computed using a softmax function with a temperature parameter ε, which adjusts the sharpness of the weight distribution. This mechanism ensures that the samples that are more closely aligned with the prototype

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Algorithm 1 Prototype Rectified Module

Require: Query set features $\{x_j^q\}_{j=1}^{N*Q}$, Original support prototypes P_n^O, Total number of classes N, Number of high-confidence query samples Z^{top}, Feature extractor parameters $F_\theta(\cdot)$.

for $n = 1$ to N **do**
  Step 1: High confidence query sample selection
  $\{d_m\} \leftarrow D(F_\theta(x_j^q), P_n^O)$
  Sort $\{d_m\}$ in ascending order and select the top Z^{top} query set samples
  Step 2: Enhanced support set sampling
  The enhanced support set $X'_{i,n}$ obtained from the Z^{top} high-confidence query set samples and augmented support set samples x_i^{aug}
  Step 3: Relationship weight calculation
  The relationship weight $\omega_{i,n}$ is calculated by the enhanced support set $X'_{i,n}$ with original support prototypes P_n^O as in Eq. (5)
  Step 4: Prototype Rectified
  The refined prototype P_n^R is established through relationship weight $\omega_{i,n}$ with enhanced support set $X'_{i,n}$ by Eq. (6)
  return P_n^R
end for

Ensure: Refined prototype P_n^R

contribute more significantly, while those that are less aligned have lower weights. The workflow of the PRM algorithm is outlined in Algorithm 1.

E. Loss Function

The final loss function is combined by cross-entropy loss L_{CE} and sample normalizer loss L_{SN} as the following equation:

$$L_{\text{total}} = L_{\text{C E}} + \lambda L_{\text{S N}}. \tag{7}$$

The hyperparameter λ is used to regulate the balance between L_{CE} and L_{SN}. L_{CE} is applied to train DiffPR-Net with hybrid support set, which is expressed as the following equation:

$$L_{CE} = -\frac{1}{M}\sum_{i}\sum_{c=1}^{N} y_{ic}\log(P(x_i^{\text{aug}}, c)) \tag{8}$$

where M refers to the sample count, while N represents the number of classes. The true label is denoted by y_{ic}, with c representing the specific class. $P(x_i^{\text{aug}}, c)$ represents the probability assigned by the model that the label of sample x_i^{aug} belongs to class c.

To address the potential generation of ambiguous samples with unknown classes by the diffusion module when adding random Gaussian noise, we have introduced the sample normalizer loss L_{SN}. This loss aims to encourage the support set samples generated by the diffusion module to be close to their respective class prototypes while being distant from prototypes of different classes, as in the following equation:

$$L_{\text{S N}} = \frac{\sum_{n=1}^{N}\sum_{i=1}^{M}||P_n - F_\theta(x_{n,i}^{\text{aug}})||_2}{\sum_{i=1}^{N}\sum_{j=i+1}^{N}||P_i - P_j||_2} \tag{9}$$

where N represents the total number of classes, while M signifies the count of support set remote sensing samples generated for each class through the diffusion module. P_n denotes the prototype of class n, and $x_{n,i}^{\text{aug}}$ stands for the ith generated support set sample within class n.

IV. Experimental Results and Analysis

A. Datasets Description

To validate the effectiveness of our proposed approach, we conducted experiments on three publicly available datasets: NWPU-RESISC45 [3], WHU-RS19 [27], and UC Merced [28]. To ensure a fair comparison, the datasets were divided into training, validation, and test sets following the data split provided in [12] and [14].

1) NWPU-RESISC45: The NWPU-RESISC45 dataset is a remote sensing dataset released by the Northwestern Polytechnical University in 2017. The dataset contains 45 scene categories with 700 images per category, totaling 31 500 images. The size of each image is 256 × 256 pixels. In our experiments, the dataset is divided into 25, 10, and 10 categories for training, validation, and testing, respectively.

2) WHU-RS19: The WHU-RS19 dataset is a remote sensing dataset released by Wuhan University in 2012. It contains 19 different scene categories with a total of 1005 images. Each category has at least 50 remote sensing images of size 600 × 600 pixels. We divide its training set, validation set, and test set into nine categories, five categories, and five categories, respectively.

3) UC Merced: The UC Merced dataset comprises 21 distinct scene categories, each containing 100 remote sensing images with a pixel size of 256 × 256. In total, the dataset contains 2100 images. The 21 categories are divided into ten, five, and six categories for training, validation, and testing, respectively.

B. Implementation Details

The PyTorch framework was used to implement our models, and experiments were conducted on an NVIDIA A30 GPU. Stochastic gradient descent (SGD) with a momentum of 0.9, an initial learning rate of $1e^{-4}$, and weight decay of $5 \times e^{-4}$ was used to update the model parameters. The learning rate was decayed at epochs 40 and 80. All input remote sensing images were resized to 256 × 256 pixels. In accordance with the meta-learning protocol, we conducted five-way one-shot and five-way five-shot scene classification tasks, with each meta-task comprising 15 query samples. During the meta-training phase, we trained for 100 epochs, with each epoch consisting of 1000 training meta-tasks and 600 validation meta-tasks. For testing, we conducted five runs, each randomly sampling 2000 meta-tasks from the test set. The accuracy reported was the average of the 2000 meta-tasks in each run. The reported accuracy was calculated as the average of five runs, with a 95% confidence interval. The support set for each class in all experiments contained five samples, generated by the diffusion module, and the number of Z^{top} in PRM was set to 5. The hyperparameter λ in (7) was set to 0.4.

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TABLE I
FIVE-WAY CLASSIFICATION ACCURACY ON THE NWPU-RESISC45 DATASET

Method	Year	Backbone	5-way	
			1-shot	5-shot
MatchingNet [29]	2016	ConvNet	54.46±0.77	67.87±0.59
ProtoNet [21]	2017	ConvNet	50.82±0.84	74.39±0.59
MAML [30]	2017	ConvNet	37.36±0.69	45.94±0.68
RelationNet [31]	2018	ResNet-12	55.84±0.88	75.78±0.57
MetaOptNet [32]	2019	ResNet-12	62.72±0.64	80.41±0.41
DLA_MatchNet [14]	2021	ConvNet	68.80±0.70	81.63±0.46
TAE-Net [33]	2021	ResNet-12	69.13±0.83	82.37±0.52
RS-MetaNet [34]	2021	ResNet-50	52.78±0.09	71.49±0.81
SPNet [15]	2022	ResNet-18	67.84±0.87	83.94±0.50
MKN [35]	2022	ResNet-12	65.84±0.89	82.67±0.55
DANet [12]	2022	ResNet-12	74.30±0.20	87.29±0.11
SGMNet [36]	2022	ConvNet	70.40±0.83	82.32±0.53
DCN [37]	2023	ConvNet	74.40±0.78	89.22±0.41
MMML [38]	2023	ResNet-12	77.35±0.73	88.19±0.47
MLLD [39]	2023	ConvNet	65.88±0.83	82.06±0.53
DL [16]	2023	ResNet-12	74.03±0.76	88.32±0.43
HiReNet [18]	2024	ResNet-12	70.43±0.90	81.24±0.58
MSFMNet [19]	2024	ResNet-12	72.03±0.45	88.62±0.46
ACL-Net [20]	2024	ResNet-12	76.13±0.24	86.54±0.23
DiffPR-Net	Ours	ResNet-12	**81.85±0.32**	**89.41±0.20**

TABLE II
FIVE-WAY CLASSIFICATION ACCURACY ON THE UC MERCED DATASET

Method	Year	Backbone	5-way	
			1-shot	5-shot
MatchingNet [29]	2016	ConvNet	46.16±0.71	66.73±0.56
ProtoNet [21]	2017	ConvNet	52.62±0.70	65.93±0.57
MAML [30]	2017	ConvNet	43.65±0.68	58.43±0.64
RelationNet [31]	2018	ConvNet	48.08±1.67	61.88±0.50
DLA_MatchNet [14]	2021	ConvNet	53.76±0.62	63.01±0.51
TAE-Net [33]	2021	ResNet-12	60.21±0.72	77.44±0.51
RS-MetaNet [34]	2021	ResNet-50	57.23±0.56	76.08±0.28
SPNet [15]	2022	ResNet-18	57.64±0.73	73.52±0.51
MKN [35]	2022	ResNet-12	58.45±0.54	77.92±0.32
DANet [12]	2022	ResNet-12	61.60±0.18	78.02±0.16
SGMNet [36]	2022	ConvNet	60.52±0.74	73.42±0.49
DCN [37]	2023	ConvNet	58.64±0.71	76.61±0.49
MMML [38]	2023	ResNet-12	61.42±0.34	79.95±0.16
MLLD [39]	2023	ConvNet	59.05±0.75	77.76±0.52
DL [16]	2023	ResNet-12	65.44±0.72	82.63±0.45
HiReNet [18]	2024	ResNet-12	58.60±0.80	76.84±0.56
MSFMNet [19]	2024	ResNet-12	62.85±0.32	80.88±0.35
ACL-Net [20]	2024	ResNet-12	59.74±0.46	74.89±0.29
DiffPR-Net	Ours	ResNet-12	**85.36±0.24**	**89.43±0.19**

TABLE III
FIVE-WAY CLASSIFICATION ACCURACY ON THE WHU-RS19 DATASET

Method	Year	Backbone	5-way	
			1-shot	5-shot
MatchingNet [29]	2016	ConvNet	60.60±0.68	82.99±0.40
ProtoNet [21]	2017	ConvNet	70.88±0.65	85.62±0.33
MAML [30]	2017	ConvNet	46.72±0.55	79.88±0.41
RelationNet [31]	2018	ConvNet	60.92±1.86	79.75±1.19
MetaOptNet [32]	2019	ResNet-12	62.05±0.76	82.60±0.46
DLA_MatchNet [14]	2021	ConvNet	68.27±1.83	79.89±0.33
TAE-Net [33]	2021	ResNet-12	73.67±0.74	88.95±0.52
SPNet [15]	2022	ResNet-18	81.06±0.60	88.04±0.28
DANet [12]	2022	ResNet-12	75.02±0.16	89.21±0.07
SGMNet [36]	2022	ConvNet	85.06±0.55	90.12±0.27
DCN [37]	2023	ConvNet	81.74±0.55	91.67±0.25
MMML [38]	2023	ResNet-12	77.76±0.69	88.40±0.54
MLLD [39]	2023	ConvNet	76.07±0.65	90.69±0.27
DL [16]	2023	ResNet-12	82.57±0.58	93.33±0.23
MSFMNet [19]	2024	ResNet-12	78.24±0.42	88.16±0.58
ACL-Net [20]	2024	ResNet-12	78.30±0.32	90.43±0.15
DiffPR-Net	Ours	ResNet-12	**85.43±0.21**	**95.69±0.10**

C. Comparison Results With Other Methods

In order to comprehensively validate the efficacy of our DiffPR-Net, we conduct comparative analyses against five representative FSL methods such as ProtoNet [21], MatchingNet [29], MAML [30], RelationNet [31], and MetaOptNet [32]; and 13 state-of-the-art (SOTA) methods in FSRSSC DANet [12] such as DLA_MatchNet [14], MKN [35], SPNet [15], MSFMNet [19], ACL-Net [20], DL [16], HiReNet [18], TAE-Net [33], RS-MetaNet [34], SGMNet [36], DCN [37], MMML [38], and MLLD [39]. The outcomes of the five typical FSL methods were extracted from the DL [16] and MLLD [39], while the performance metrics of the typical FSRSSC methods were obtained from their original publications. The experimental comparison results on the NWPU-RESISC45, UC Merced, and WHU-RS19 datasets are shown as Tables I–III, respectively. The optimal results are emphasized in bold, while the second-best results are indicated with an underline each.

It can be observed from Table I that the DiffPR-Net method achieves the best performance in both the five-way one-shot and five-way five-shot settings on the NWPU-RESISC45 dataset. In the five-way one-shot setting, the DiffPR-Net method demonstrated superior performance to the second-best methods, MMML [38] and ACL-Net [20], with an improvement of 4.5% and 5.72%, respectively. In the five-shot setting, the DiffPR-Net method achieved a classification accuracy of 89.41%, outperforming the second-best methods, DCN [37], DL [16], and MMML [38], by 0.19%, 1.09%, and 1.22%, respectively.

As evidenced by Tables II and III, the DiffPR-Net demonstrates optimal performance on both the UC Merced and WHU-RS19 datasets. In the five-way one-shot setting on the UC Merced dataset, our method outperforms the second-best method DL [16] by 19.92%. In the five-shot setting, the performance improvement compared to DL [16] is 6.8%. The remote sensing images in the UC Merced dataset display significant intraclass similarity. The diffusion module generates samples that introduce image diversity, thereby expanding the representation space of the original dataset. Generating more diverse samples under the one-shot setting enhances the model's generalization ability, enabling it to better adapt to new samples and significantly improve classification accuracy. The results of our DiffPR-Net method on the WHU-RS19 dataset show that our one-shot performance exceeds that of the second-best method SGMNet [36] by 0.37%. In the five-shot setting, our method outperforms DL [16] by 2.36%. These results demonstrate the effectiveness of our method in improving the performance of FSRSSC.

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D. Ablation Study

To ascertain the effectiveness of the individual components within our DiffPR-Net, ablation experiments are conducted in both five-way one-shot and five-way five-shot settings on three distinct datasets, as is presented in Table IV. The baseline configuration refers to the Meta DeepBDC [40] method, which applies ResNet-12 as the feature extractor. The results of baseline on the NWPU-RESISC45, UC Merced, and WHU-RS19 datasets are shown as the first row of Table IV.

1) Ablation on DA: As is shown in the second row of Table IV, the classification accuracy of the three datasets improved by 9.88%, 9.55%, and 1.37% in the one-shot setting, which demonstrates the effectiveness of enriching sample diversity and enhancing the generalization capabilities in data-scarce environments. For the five-shot tasks, the classification accuracy improved by 5.2%, 2.42%, and 2.37%, respectively. The DA module is incorporated into the network to augment the support set samples in each meta-task, resulting in improved classification accuracy for each dataset.

To evaluate the DA capabilities of our DiffPR-Net in comparison to GAN-based methods, we compare the mutual information (MI), cosine similarity (CS), inception score (IS), and parameter numbers (Params) with MARTA GANs [41] and MF-WGANs [42]. The comparison results are presented in Table V, where the results of MARTA GANs [41] and MF-WGANs [42] are from their original articles. The proposed method outperforms GAN-based methods in all metrics. However, the parameter number of the DiffPR-Net is slightly higher than MARTA GANs and MF-WGANs, which reflects its robust learning capabilities and indicates a moderately computational resource during both inference and training. While the elevated computational complexity may enhance model performance, it could also impact its practical applicability in resource-constrained environments. Therefore, it is essential to consider appropriate hardware support and optimization strategies.

Furthermore, we assessed the efficacy of samples generated by the diffusion module in comparison to real samples. We conducted experiments on three datasets, considering three settings: five-way one-shot, five-way six-shot, and one-shot+five generated samples. The results presented in Fig. 6 demonstrate that the one-shot+five generated samples setting is comparable to six-shot on the NWPU and UCML datasets.

To compare the visual expression of generated images by our proposed DiffPR-Net and GANs, the examples of partially random are shown as Fig. 7. While GANs demonstrate strong performance in general image generation tasks, it often exhibits higher levels of noise and artifacts in remote sensing images, negatively impacting classification performance. In contrast, our DA produces more stable and detail-rich images, enhancing accuracy in classification tasks.

2) Ablation on DAFM: From the third row of Table IV, the results with DAFM increase of 0.82%, 2.6%, and 2.56% in the NWPU, UCML, and WHU datasets of the one-shot setting. The corresponding accuracy enhancements for the five-shot tasks were 2.44%, 2.73%, and 1.51%, which demonstrates that the proposed DAFM effectively guides the network to capture rich feature information in remote sensing scene images and to focus on key areas. The efficacy of the DAFM was further validated through visualization using gradient-weighted class

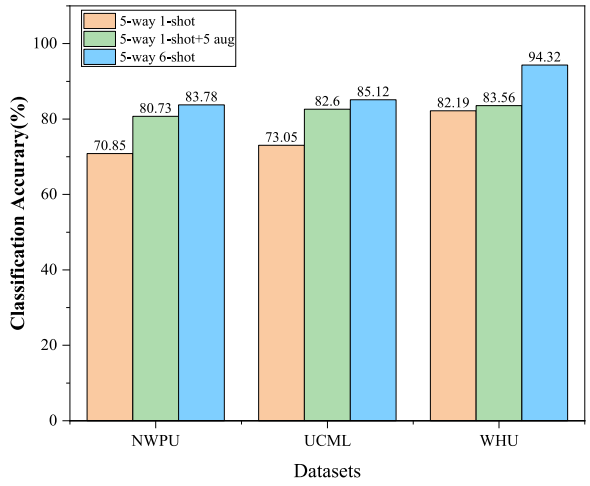


Fig. 6. Performance comparison of generated samples with real samples.

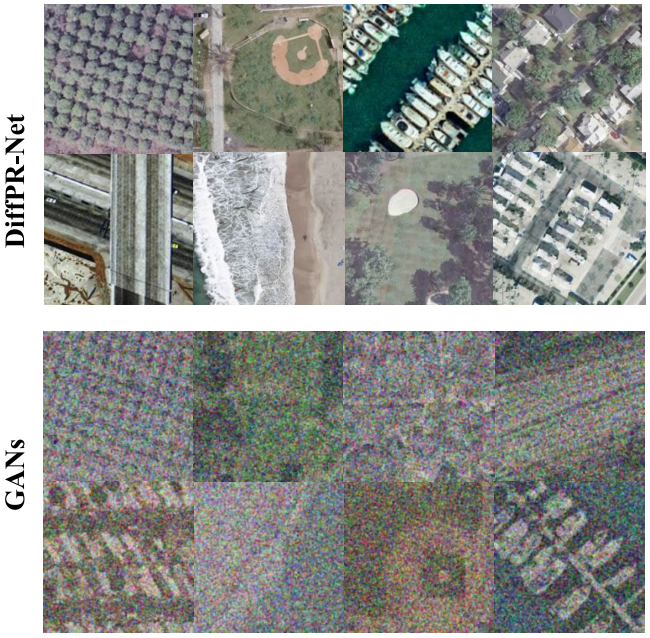


Fig. 7. Examples of partially randomly generated images by our proposed DiffPR-Net and GANs.

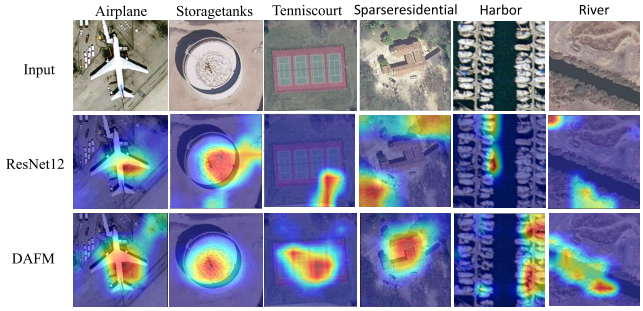


Fig. 8. Grad-CAM visualization map. Each column contains the input image, the attention map of baseline model (ResNet-12) and that of the model with DAFM (ResNet-12 + DAFM).

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TABLE IV
RESULTS OF ABLATION STUDIES FOR THE THREE DATASETS

DA	DAFM	PRM	5-way					
			NWPU-RESISC45		UCML		WHU	
			1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
×	×	×	70.85	83.12	73.05	84.64	82.19	93.04
✓	×	×	80.73	88.32	82.60	87.06	83.56	95.41
×	✓	×	71.67	85.56	75.65	87.37	84.75	94.55
×	×	✓	72.74	86.74	74.44	88.07	84.33	95.08
×	✓	✓	74.31	87.05	77.09	88.45	84.83	95.10
✓	✓	✓	81.85	89.41	85.36	89.43	85.43	95.69

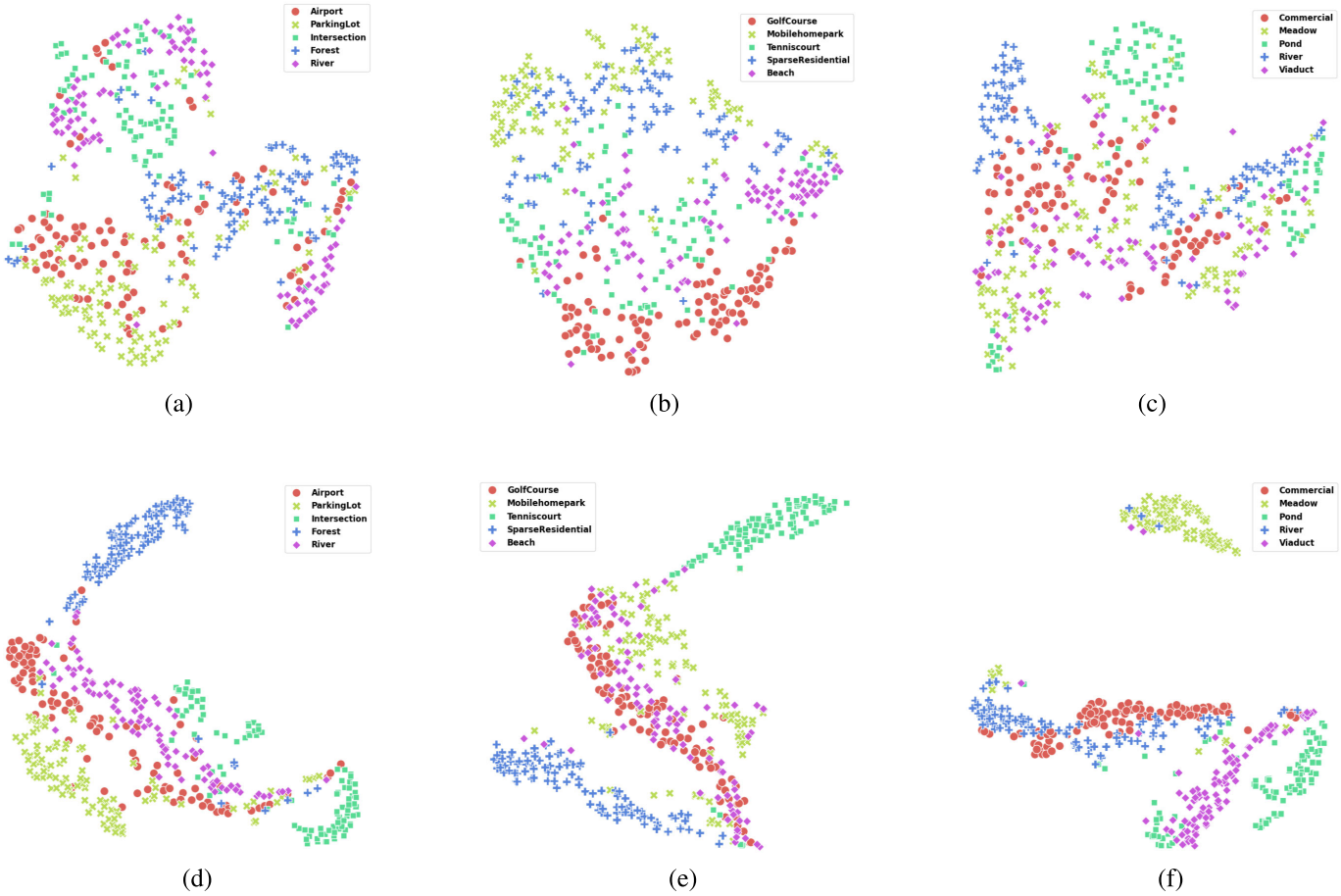


Fig. 9. t-SNE visualization analysis was conducted to examine the influence of PRM on intraclass compactness and interclass dispersion of deep features in ProtoNet and DiffPR-Net on various remote sensing datasets. (a) ProtoNet on NWPU. (b) ProtoNet on UCML. (c) ProtoNet on WHU. (d) DiffPR-Net on NWPU. (e) DiffPR-Net on UCML. (f) DiffPR-Net on WHU.

TABLE V
COMPARISON OF DATA AUGMENTATION BETWEEN GANS AND OUR PROPOSED DIFFPR-NET

Method	MI	CS	IS	Params
MARTA GANs [41]	0.5440	0.7263	6.21	2.8
MF-WGANs [42]	0.7183	0.8682	11.60	4.2
DiffPR-Net	0.8213	0.9092	13.54	6.3

TABLE VI
COMPARISON OF INTERCLASS SEPARATION AND INTRACLASS COMPACTNESS

Method	NWPU-RESISC45		UCML		WHU	
	R_{inter}	R_{intra}	R_{inter}	R_{intra}	R_{inter}	R_{intra}
ProtoNet	48.63	57.29	53.32	58.89	77.18	58.45
DiffPR-Net	65.35	43.84	62.16	42.18	84.23	46.37

activation mapping (Grad-CAM) [43], as shown in Fig. 8. The visualizations indicate that the activation regions of the baseline model contain redundant information, which hinders effective remote sensing feature extraction. However, the introduction of the DAFM results in notable improvements. Grad-CAM produces heatmaps that more accurately and significantly emphasize key areas relevant to the target category.

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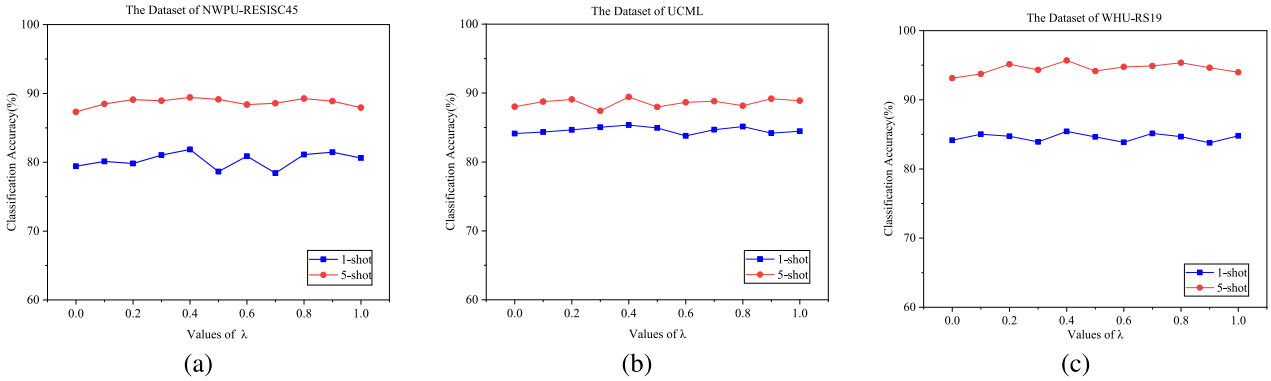


Fig. 10. Impact of weight coefficient λ on three datasets. Results on (a) NWPU-RESISC45 dataset, (b) WHU-RS19 dataset, and (c) UC Merced dataset.

TABLE VII
RESEARCH ON THE IMPACT OF z^{TOP} QUANTITY VARIATION

Number of Z^{top}	5-way 1-shot	5-way 5-shot
0	70.85±0.43	83.12±0.23
5	72.74±0.46	86.74±0.23
10	71.32±0.82	84.61±0.69
15	67.05±0.93	81.17±0.78

3) Ablation on Prototype Rectification Module: The integration of the PRM into the prototype network represents a critical advancement in enhancing the robustness and precision of scene classification tasks within remote sensing applications, as is shown in the fourth row of Table IV. The data clearly demonstrates that the PRM effectively mitigates prototype bias, resulting in more coherent feature representations and improved classification outcomes. In order to demonstrate the efficacy of the PRM designed to address prototype bias, Fig. 9 presents t-SNE [44] visualization results for the one-shot tasks of three remote sensing datasets, comparing the prototype network with the PRM-integrated network. Due to the experimental setup (five-way), only five classes were selected from the test set for the purpose of displaying the distribution of features. It can be observed that the class features obtained by the prototype network are disordered and fail to provide valuable information for scene classification. The PRM enhances the compactness of deep features within a class and the dispersion between classes, thereby improving classification accuracy. The efficacy of the PRM proposed in this study is supported by empirical evidence, as illustrated in Fig. 9.

The quantitative comparison of interclass separation R_{inter} and intraclass compactness R_{intra} between ProtoNet and DiffPR-Net is shown as Table VI, which reveals that DiffPR-Net outperforms ProtoNet in both metrics. DiffPR-Net demonstrates a significantly higher interclass separation compared to ProtoNet, indicating its superior capability to maintain greater differentiation between distinct classes. Additionally, DiffPR-Net achieves lower intraclass compactness than ProtoNet, which shows that it facilitates tighter clustering of samples within the same class. This improvement in interclass separation, combined with enhanced intraclass compactness, allows DiffPR-Net to gain a notable advantage in FSRSSC, effectively enhancing the model's discriminative power and generalization performance.

E. Parameter Analysis

To examine the effect of the balance hyperparameter λ in (7) on the classification performance of proposed DiffPR-Net, we conduct experiments on three datasets. We evaluated the model's performance in the five-way one-shot and five-way five-shot testing classification tasks by varying λ from 0 to 1 to determine the optimal value for λ. As illustrated in Fig. 10, the optimal performance was achieved when λ was set to 0.4 in both the one-shot and five-shot tasks.

Experiments are conducted on the NWPU dataset to investigate the influence of different values of Z^{top} on the performance of PRM. The results, as shown in Table VII, indicate that an increase in the value of Z^{top} from 0 to 5 led to a gradual improvement in PRM's performance. However, a further increase in Z^{top} to 15 resulted in a decline in performance. An appropriate number of Z^{top} samples can effectively correct the initial prototypes, improving their accuracy. However, an excessive number of Z^{top} samples can disrupt the consistency of the prototypes and introduce noise. This analysis provides valuable insights for optimizing and refining the PRM model.

To analyze impact on the number of generated samples, we conduct experiments of optimal sample count, as is illustrated in Fig. 11. It demonstrates that performance improves as the number of generated samples increases, but only up to a certain point. Specifically, when the number of generated samples is set to 5, accuracy reaches its peak. This deterioration is likely due to the model being overwhelmed by excessive synthetic data, which may introduce noise or redundant information, leading to overfitting or confusion in the feature space. Therefore, to balance between data diversity and relevance, the generated samples are set as 5.

F. Backbone Architectures Analysis

To investigate the impact of different backbone architectures, we perform comparative experiments on four distinct backbones: ConvNet4-64, ConvNet4-256, ResNet-18, and ResNet-12, which are selected for varying complexities

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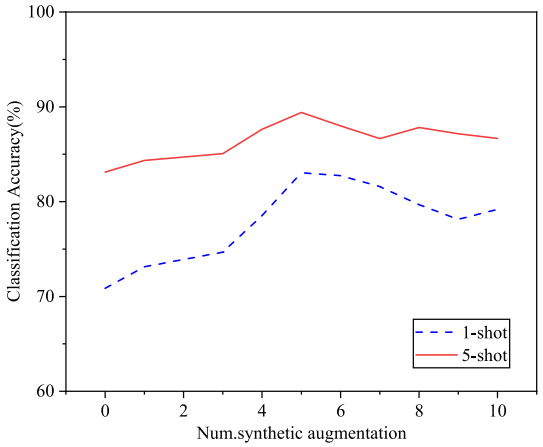


Fig. 11. Performance variation with different numbers of generated samples for NWPU-RESISC45.

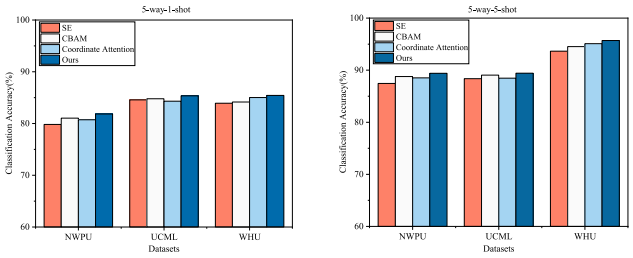


Fig. 12. Classification accuracy of different attention modules on the NWPU-RESISC45, WHU-RS19, and UC Merced datasets.

TABLE VIII
CLASSIFICATION ACCURACY OF DIFFERENT BACKBONES ACROSS THREE DATASETS

Backbone	NWPU-RESISC45		UCML		WHU	
	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
ConvNet4-64	71.32	82.33	79.16	81.19	81.32	91.04
ConvNet4-256	74.24	84.47	82.73	85.75	82.16	93.15
ResNet-18	77.31	86.42	83.33	86.48	83.47	92.93
ResNet-12	**81.85**	**89.41**	**85.36**	**89.43**	**85.43**	**95.69**

and feature extraction capacities to facilitate a comprehensive performance evaluation across different datasets. Both ConvNet4-64 and ConvNet4-256 consist of four convolutional blocks, each comprising a 3 × 3 convolution layer, a batch normalization layer, a ReLU activation layer, and a 2 × 2 max pooling layer, resulting in 64-D and 256-D feature vectors, respectively. In contrast, the ResNet architectures employ residual connections to mitigate the vanishing gradient problem, allowing for the training of deeper networks. Specifically, ResNet-12 includes 12 layers with skip connections that enhance gradient flow during backpropagation, improving learning efficiency and model performance. The comparative results of four backbones on three datasets are shown as Table VIII, where the ResNet-12 achieves the greatest performance. Additionally, it is noted that overly complex backbone networks are prone to overfitting, which can negatively impact overall performance.

G. Comparison With Other Attention Modules

To showcase the effectiveness of our proposed DAFM in feature extraction, we conducted a comparative analysis with prevalent attention mechanisms such as SE [45], convolutional block attention module (CBAM) [46], and coordinate attention [47]. The classification accuracies achieved by various attention modules across the three remote sensing datasets are shown as Fig. 12. Our DAFM outperformed other attention mechanisms consistently in both one-shot and five-shot scenarios. Unlike SE, which focuses solely on channel-wise information, and CBAM, which applies channel and spatial attention sequentially, our DAFM transmits highly fused image detail features from higher layers to lower layers, allowing the model to focus more effectively on discriminative regions. Compared to coordinate attention, which captures long-range dependencies, DAFM excels in preserving fine-grained image details, enhancing both local and global feature integration. This results in improved performance in complex remote sensing scene classification.

V. CONCLUSION

This study presents the DiffPR-Net to address few-shot challenges in remote sensing scene classification. To overcome the scarcity of samples, we introduce the DA to augment support set samples within meta-tasks, thereby generating high-fidelity remote sensing images. The DAFM mechanism is designed to combine global and local detailed features, allowing for inter-layer transmission to extract powerful and distinctive features. To refine prototypes and reduce prototype bias, we enhance the support set using a pseudo-labeling strategy and assign temporary labels to unlabeled data based on adaptive prediction confidence. The proposed DiffPR-Net corrects initial prototypes by dynamically assigning relationship weights. This is achieved through the weighted sum of support samples and pseudo-labeled query samples. The PRM enhances category feature representation by emphasizing intraclass and interclass relationships among small samples. The experimental results on three benchmark remote sensing scene classification datasets validate the effectiveness of the proposed DiffPR-Net, showcasing competitive performance. While DiffPR-Net demonstrates superior performance in FSRSSC, it is crucial to address the challenges posed by high computational demands. We will focus on optimizing the computational efficiency, incorporating strategies such as model compression, lightweight architecture design, and adaptive computation techniques. It aims to achieve an effective balance between performance and resource utilization, thereby enhancing the feasibility and applicability of the model in real-world scenarios.

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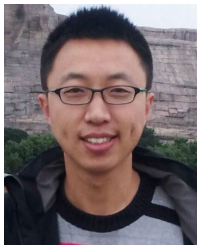
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